

Immersive Audio

Learn what's special about Dolby Atmos for Music, a new 3D audio format

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Q First off, what is Dolby Atmos for Music? How does it make a difference to the user's listening experience?

Mr Pankaj Kedia: Dolby Atmos for Music is a new way of creating music and a new way of experiencing music. It reveals new details in the music that we have not heard. It might be certain instruments, certain riffs, certain percussion beats, or even just the singer breathing into the mic. There are details that we've not heard in songs, they will be revealed. The second thing is that there is an incredible amount of clarity, detail and separation between the different elements of music. It's a much richer experience, and a much stronger emotional connect. It tends to bring the audience or the fan closer to the artist. The sense a user gets is as if you're sitting in the studio that the song was being recorded.

Q How does Dolby Atmos for Music work technically?

Mr Pankaj Kedia: So, we'll talk about the creation bit and the experience bit. From a creation perspective, what we do with Dolby Atmos Music is that we provide new tools and new techniques to the people who create the music including the artist, the producer, the mixing engineer. With these new tools and techniques, it takes away a lot of limitations that are present in creating music currently. Essentially music today exists in stereo, which is two channels of sound. There are a lot of significant limitations and constraints pertaining to what you can do with it. For example, if you want the vocals to be



Mr Pankaj Kedia,
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heard, you can't really make instruments stand out because the vocals would get drowned. In Dolby Atmos for Music, they're creating music in a completely three-dimensional sphere. It's actually a room that they create in. They can then place instruments and sounds as individual objects around the listener and they can move them around very precisely as well. So not only is the placement precise, but the movement is also precise. This does mean that for people who are creating the music, if they're doing it in a studio, there are some additional speakers required, but if they're doing it on headphones, they can even mix it on headphones. So, there are various ways of creating the music.

Now, coming to the experience part, from a consumer perspective, one of the most common use cases

of listening to music is with a mobile device using headphones. From a mobile perspective, Dolby Atmos Music works with any smartphone that is Atmos capable and with any pair of headphones. Today, almost every leading manufacturer includes Atmos capabilities on their smartphones. Apple has it on all their smartphones. It's widely deployed across Samsung's portfolio. You also have Xiaomi, Oppo, Huawei, OnePlus that include Dolby Atmos capabilities. Globally, there are more than a million phones being shipped every day with Dolby Atmos capabilities. If a song is created in Atmos and it is available on a streaming service with Atmos, you can then play it back on an Atmos mobile by either using the onboard speakers or your favourite pair of headphones. What you will experience is not just sound coming from two point sources, but from all around you. It will immerse you in the music like a song bath happening all around you. From a technical perspective, it widens what we call the soundstage, so it's a much bigger soundstage you're hearing. You know the direction the sound is coming from as if you were sitting in the middle and the band is playing around you. The most frequent feedback we hear is that it is the new way to listen to music and you can never go back to stereo.

Q Dolby Atmos Music was showcased at CES 2020, since then what are some of the improvements that have been made to the technology?

Mr Pankaj Kedia: Dolby is a company that has innovation at its core, so we're constantly inventing and